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AN ATTENTION ECONOMY MODEL OF CO-EVOLUTION BETWEEN CONTENT QUALITY AND AUDIENCE SELECTIVITY

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ABSTRACT

Human attention has become a scarce and strategically contested resource in digital environments. Content providers increasingly engage in excessive competition for visibility, often prioritizing attention-grabbing tactics over substantive quality. Despite extensive empirical evidence, however, there is a lack of theoretical models that explain the fundamental dynamics of the attention economy. Here, we develop a minimal mathematical framework to explain how content quality and audience attention coevolve under limited attention capacity. Using an evolutionary game approach, we model strategic feedback between providers, who decide how much effort to invest in production, and consumers, who choose whether to search selectively for high-quality content or to engage passively. Analytical and numerical results reveal three characteristic regimes of content dynamics: collapse, boundary, and coexistence. The transitions between these regimes depend on how effectively audiences can distinguish content quality. When audience discriminability is weak, both selective attention and high-quality production vanish, leading to informational collapse. When discriminability is sufficient and incentives are well aligned, high- and low-quality content dynamically coexist through feedback between audience selectivity and providers' effort. These findings identify two key conditions for sustaining a healthy information ecosystem: adequate discriminability among audiences and sufficient incentives for high-effort creation. The model provides a theoretical foundation for understanding how institutional and platform designs can prevent the degradation of content quality in the attention economy.

***K*eywords** Attention economy · Adaptive feedback · Replicator dynamics

Introduction

In today's digital society, the attention economy has become a central paradigm for understanding online information ecosystems. Human attention functions as a limited resource subject to intense competition [1, 2, 3, 4, 5, 6, 7]. This competition unfolds across digital media spaces, including news and social media platforms, video-sharing sites, and live-streaming environments [8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19], where user-generated and professional content

continually vie for visibility. As digital platforms lower the barriers to content creation, the volume of online content has surged. This expansion has intensified competition for user attention and enabled even low-quality material to spread widely [20, 21]. This imbalance between abundant content and users' limited capacity to process it highlights the fundamental scarcity of attention in online environments.

A widely observed pattern in such competitive environments is the proliferation of low-effort content, including clickbait, optimized to exploit platform algorithms and maximize short-term engagement [22, 23, 24]. Economic models show that competition for fragmented attention can depress information quality [25] and may even incentivize misinformation when short-term gains outweigh long-term credibility [26]. Recent advances in generative AI have amplified these pressures. They enable the rapid production of large volumes of text, images, and videos at minimal cost [27, 28, 29]. Although these strategies may be economically rational for individual providers, they collectively threaten the long-term viability of high-quality content and the informational health of online ecosystems. These observations raise a fundamental question: under what conditions can high-quality content remain sustainable in environments where limited attention is strategically distributed?

To address this question, it is essential to recognize that online information environments are shaped by the strategic adaptations of both content providers and consumers. Providers adjust their production strategies in response to competitive pressures and anticipated visibility. Consumers allocate their attention based on perceived credibility and content quality. These reciprocal adjustments create feedback loops that shape the evolving relationship between content production and attention allocation [30, 31, 32]. Previous studies have largely examined either collective attention patterns or individual attention decay [33, 34, 9, 35]. However, these approaches do not capture how providers and consumers jointly adapt under attention constraints. A theoretical framework that integrates their coadaptive behaviors and clarifies how such constraints shape long-term evolutionary outcomes is still lacking. In particular, many existing models treat attention allocation and content quality as exogenous, which limits their ability to capture the feedback between the two. As a result, how limited attention shapes the coevolution of quality provision and consumer behavior remains insufficiently understood.

Building on these observations, we develop a theoretical framework that explains when high-quality content can persist under limited attention. Unlike previous approaches that examine attention flows or individual behavior in isolation, our model jointly captures the strategic adaptation of content providers and consumers under explicit attention constraints. We construct a two-population evolutionary game with behavioral microfoundations, in which interactions under finite attention capacity generate a piecewise-smooth replicator system derived through a mean-field approximation. This formulation enables us to analytically identify the parameter thresholds that separate collapse, boundary, and coexistence regimes, providing a unified account of how attention constraints shape the long-term evolutionary dynamics of content quality.

Our analysis reveals that limited attention induces a fundamental divide in content dynamics: without sufficient discriminability, systems generically collapse into low-quality convergence, whereas boundary outcomes and stable coexistence arise only under specific combinations of attention capacity and providers' incentive strength. These evolutionary patterns clarify when high-effort content can persist in attention-driven environments and offer theoretical guidance for designing platforms that maintain informational quality. We formalize these mechanisms by introducing the evolutionary framework on which our analysis is based.

Results

Attention Economy Model

To formalize the coadaptive mechanisms discussed above, we construct a minimal evolutionary framework that captures how content quality and audience attention coevolve under limited cognitive resources. Digital environments are characterized by intense competition for visibility, which makes the incentives of providers and consumers inherently interdependent. The structure of the model is summarized in Fig. 1.

Content providers choose between two strategies. Under the high-effort strategy H, they produce high-quality content at a cost $c_H > 0$. Under the low-effort strategy L, they generate lower-quality content with no additional cost. We consider a fixed population of m providers, each producing one item of content per period, regardless of strategy. Providers receive a reward $r > 0$ per view, so the payoff advantage of high-effort production depends on how consumers allocate attention across available items. The reward r is identical for high- and low-quality items, reflecting platform environments, such as video-sharing platforms, where provider revenue is largely determined by view counts rather than intrinsic content quality.

Consumers also choose between two strategies. Under the active strategy A, they search for and selectively consume high-quality items while paying a per-period cost $c_A > 0$ that represents search and evaluation effort. Under the passive

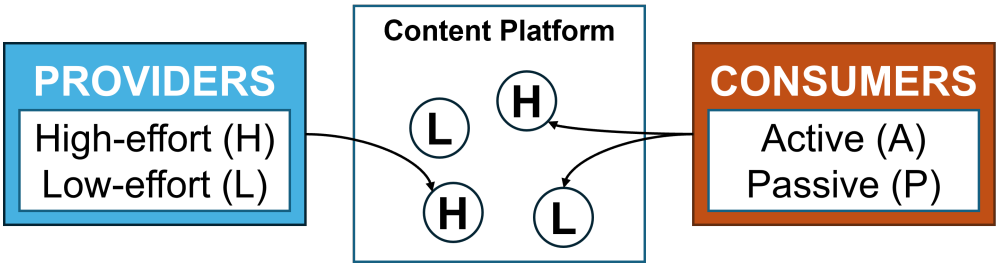

Figure 1: Conceptual diagram of the attention economy model. Providers invest high (H) or low (L) effort in content creation, while consumers allocate limited attention (σ) through active (A) or passive (P) viewing. Their strategic feedback determines the evolution of content quality and attention distribution.

Table 1: Summary of model parameters.

Symbol	Description
m	Number of content providers
n	Number of content consumers
σ	Attention capacity (fraction of attendable items)
b_H	Benefit obtained from a high-quality item
b_L	Benefit obtained from a low-quality item
c_H	Cost of adopting the high-effort strategy
c_A	Cost of adopting the active viewing strategy
r	Reward per view received by providers
$x(t)$	Fraction of high-effort providers at time t
$y(t)$	Fraction of active consumers at time t

strategy P, they allocate their attention budget uniformly across all items produced. We consider a fixed population of n consumers, each with a limited attention capacity $\sigma \in (0, 1]$, which allows them to consume only a fraction σ of items produced in each period. Consumer benefits from each item are additively separable.

Attention is treated as a scarce cognitive resource. The parameter σ formalizes this constraint by limiting each consumer's attention budget, forcing providers to compete for inclusion within a narrow attention window. As a result, visibility becomes a key determinant of provider incentives and directly reflects consumer selection behavior. A key implication of this assumption is that attention scarcity enters payoffs directly, creating a qualitative shift in incentives depending on whether high-quality supply is sufficient to saturate consumers' attention budgets. The boundary $x = \sigma$ thus marks the point at which active consumers can fully satisfy their attention budgets using only high-quality items, eliminating any residual allocation to low-quality content and altering the payoff structure accordingly.

Given these incentives, the composition of strategies in each population evolves over time. Strategies adjust in proportion to their payoff differences, with higher-payoff strategies becoming more common. This replicator-like adjustment creates feedback between content quality and consumer selectivity.

Interactions in each period proceed as follows:

1. Providers produce one item according to their chosen strategy.
2. Consumers select $m\sigma$ items. Active consumers allocate their attention entirely to high-quality items when these are available in sufficient quantity. When the supply of high-quality items is insufficient, they consume all available high-quality items and then randomize over low-quality items to fill their remaining attention slots. Passive consumers sample uniformly from all items.
3. Providers receive reward r per view.
4. Consumers obtain benefit b_H from high-quality and b_L from low-quality items, with $b_H > b_L > 0$.

The model parameters are listed in Table 1.

Let $x(t)$ denote the fraction of high-effort providers and $y(t)$ the fraction of active consumers at time t. These state variables summarize the population-level state. We capture the resulting feedback through a two-population replicator framework, with the full mathematical formulation provided in the Materials and Methods section.